When to Raise Taxes:
State Dependence in the UK Tax Multiplier

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Abstract

A tax rise costs about twice as much when the economy has slack. Using UK narrative tax measures from 1918 to 2018 and quarterly household consumption from 1955, I estimate the consumption response to tax changes that were not anticipated, and split it by whether unemployment was above or below its own recent average when the change took effect. Against a tax rise worth one per cent of annual GDP, consumption falls by 7.3 per cent at its trough when there is slack and by 3.2 per cent when there is not. The difference is estimated directly, with its own standard error, and is significant at eight of seventeen horizons. Consumption is the better outcome to use: it responds more than output, on three times as many horizons, and unlike the output response it survives the removal of the single Budget on which the published UK multiplier rests. The shock series reproduces Cloyne (2013) exactly before being extended, which fixes the conventions the estimate turns on. Three further questions are posed and left unanswered because the record cannot carry them: which tax matters, whether pre-announcement blunts the effect, and who bears the cost. Each is reported as a measured limit rather than a caveat.

Keywords: fiscal multipliers, narrative tax shocks, state dependence, local projections, household consumption.
JEL classification: E32, E62, H20, H30.

1 Introduction

Governments choose when to raise taxes. The advice they receive rarely turns on that choice, because the empirical literature mostly reports a single number: the average response of output to a tax change, estimated over whatever sample is available. If the response is the same whenever the change lands, the timing is a political question and not an economic one.

This paper asks whether it is the same, and finds that it is not.

The exercise is possible because of a long narrative record. Romer and Romer (2010) identify legislated US tax changes from the contemporaneous documents and code the motivation behind

*Working paper, incomplete by design. Section 6 sets out the part of the project that is not finished and says what it needs. Everything else reproduces end to end from the replication package. I thank James Cloyne for making the underlying narrative dataset public, and Cloyne, Dimsdale, Postel-Vinay and Hürtgen for depositing the interwar measures and the macroeconomic panel used here, without which this paper could not exist.

each, so that changes made for reasons unrelated to the state of the economy can be separated from those made in response to it. Cloyne (2013) builds the corresponding UK account for 1945 to 2009, and it has since been extended back into the interwar period by Cloyne et al. (2023) and forward to 2020 by Hürtgen et al. (2024), with the combined series used at century length by Cloyne et al. (2025). A companion paper (Conway, 2026) extends the measure-level record from Budget 2004 to Autumn 2018 and shows that the interval between announcing a UK tax change and its taking effect has roughly quadrupled since 1945, so that the share of the tax impulse arriving with more than four months of notice has risen from 0.18 to 0.70.

That finding shapes what can be done here. A measure announced two years before it takes effect is not a surprise on the day the money moves, and treating it as one estimates something other than a causal response. The projections below are therefore run on measures whose announcement and effect nearly coincide, which is a minority of the modern record and the price of the estimate meaning anything.

The question of whether the multiplier depends on the state of the economy is not new. Auerbach and Gorodnichenko (2012) find larger output responses to fiscal policy in US recessions, and Ramey and Zubairy (2018) dispute the finding for government spending using a longer sample and a different treatment of the news. Neither settles the question for tax changes, and neither is about the United Kingdom. What this paper adds is a UK answer, on household consumption rather than output, from a narrative series whose construction is verified rather than assumed.

The result is that against a tax rise worth one per cent of annual GDP, household consumption falls by 7.3 per cent at its trough when unemployment is above its own recent average, and by 3.2 per cent when it is not. Expressed per pound raised, the same tax rise costs roughly £4.50 of household spending in a slack economy and roughly £2 in a tight one. The gap is estimated directly rather than inferred from two separate regressions, and is significant at eight of seventeen horizons.

Three further questions are posed in Section 5 and answered in the negative. The record cannot say which tax matters, because most instruments are one or two episodes. It cannot say whether pre-announcement blunts a tax change, because the test lacks the power to detect an effect even as large as the one it is being compared against. And it cannot say who bears the cost, because splitting a measure by the incidence of its instrument produces two series that move together almost perfectly. The last of these is the subject of Section 6, which describes work in progress rather than a result.

2 Data

2.1 The narrative record

Three public sources, all carrying announcement dates, implementation dates, a revenue costing and the same motive classification, are chained into one measure-level record.

The exogeneity restriction is the identifying assumption and it is not mine: measures are used only where the original coder judged the motivation unrelated to the current state of the economy. Unlike the companion paper, which studies how long the policy process takes and

Table 1: The measure-level record

Period	Source	Datable measures	Of which exogenous
1918–1938	Cloyne et al. (2023), Zenodo, CC-BY	204	89
1945–2009	Cloyne (2013)	1,221	1,221
2004–2018	Conway (2026)	173	173

Exogenous counts are of measures usable in estimation. 1939 to 1944 is absent from every source and 1921 contains no datable measure, so a century-long specification carries holes that estimation must handle rather than patch. The interwar block carries no tax-type field and so cannot enter any instrument-level specification.

therefore needs no such restriction, a causal estimate does need it.

One design choice differs from the companion paper. That paper restricted attention to household-relevant measures, which required Cloyne's `Group` field, which his own documentation disclaims. The interwar block has no such field at all, so the restriction cannot be carried back, and dropping it also matches the construction of the published exogenous series. The household-restricted variant is built alongside and the choice does not change what follows.

2.2 Reproducing the published series before extending it

A narrative shock series is a set of conventions as much as a set of measures, and small differences in those conventions matter more than they look. Before extending anything, the construction here was made to reproduce Cloyne (2013) exactly.

His aggregation code, published with his dataset, pushes an implementation date forward by 45 days before taking its calendar quarter, replaces an implementation date that precedes its own announcement with the announcement date, treats a measure as a surprise when it takes effect within 90 days, and divides by annualised nominal GDP. Rebuilding his 1955 to 2009 surprise series from his raw file under those rules returns it at a correlation of 1.0000, with a maximum difference of zero across all 220 quarters.

That exercise then prices each convention. Table 2 switches one at a time away from his baseline and reports the correlation with his published series.

Table 2: What each dating convention is worth

Construction	Correlation with the published series
His conventions exactly	**1.0000**
A 120-day surprise threshold instead of 90	0.9236
A half-month quarter rule instead of his 45-day shift	0.9316
Retroactive dates kept rather than replaced	0.9648
A different nominal GDP vintage	0.9990
All four differences together	0.8643

The GDP vintage is irrelevant and the threshold matters most. The companion paper uses the 120-day rule for reasons specific to the British fiscal calendar and keeps it; this paper uses the 90-day convention, because comparability with the published multiplier literature is what matters here.

2.3 Outcomes

The macroeconomic panel is the one Cloyne et al. (2025) estimate on, taken from their replication deposit: quarterly real and nominal GDP, the deflator, unemployment, Bank Rate, the exchange rate, prices, revenue, expenditure and the deficit, 1918 to 2020, sourced from Thomas and Dimsdale (2017). Using their outcomes rather than assembling my own makes every comparison with the published estimates like for like. Their narrative government spending shocks, over the same span, are carried as a control.

Household consumption is not in that deposit. It is taken from Thomas and Dimsdale (2017) directly, which is the same underlying source, so consumption and output come from one vintage rather than two. Their real GDP and the panel's agree at a correlation of 1.0000 on levels and 0.9999 on growth. The expenditure components begin in 1955Q1 and version 3.1 ends in 2016Q4, so consumption work runs over 248 quarters against 1945 to 2018 for output.

2.4 Identifying variation

Table 3 reports what there is to work with. It is not abundant, and it is worth seeing before any estimate is read.

Table 3: Standard deviation of the quarterly shock, per cent of annual GDP

Era	Quarters	Unanticipated	Non-zero quarters	News
1920–38	76	0.227	14	0.012
1945–79	140	0.259	41	0.317
1980–99	80	0.158	40	0.125
2000–18	76	**0.046**	35	0.142

The surprise series keeps roughly the same number of events throughout and loses four-fifths of its amplitude. Surprises did not stop happening; they stopped being large. That is the companion paper's central finding seen from the estimation side, and it is why no result below is reported for the modern period alone.

3 Consumption, not output

For horizon h, the projection is

$$100 \times (\log y_{t+h} - \log y_{t-1}) = \alpha_h + \beta_h s_t + \Gamma_h(L) + \varepsilon_{t+h},$$

following Jordà (2005), where s_t is the unanticipated tax shock as a per cent of annual GDP, and the controls are four lags of the shock, four lags of outcome growth, and the contemporaneous narrative spending shock. Standard errors are Newey-West at bandwidth $h + 1$. A positive shock is a tax rise, so a contractionary response is $\beta_h < 0$, and β_h is the multiplier in the sense of Romer and Romer (2010).

Run on output, the estimate is respectable and fragile. Run on consumption, it is larger and holds.

Consumption is 61 per cent of GDP and moves more than output in absolute terms, so households carry more than their share of the adjustment. Figure 1 shows both paths.

Table 4: Response to an unanticipated tax rise worth one per cent of annual GDP

Outcome	Trough	Quarter	95 per cent interval	t	Significant horizons
Real GDP	-3.86	10	$[-7.04, -0.67]$	-2.37	4 of 17
Household consumption	$\mathbf{-5.31}$	9	$[-8.23, -2.40]$	$\mathbf{-3.57}$	**12 of 17**

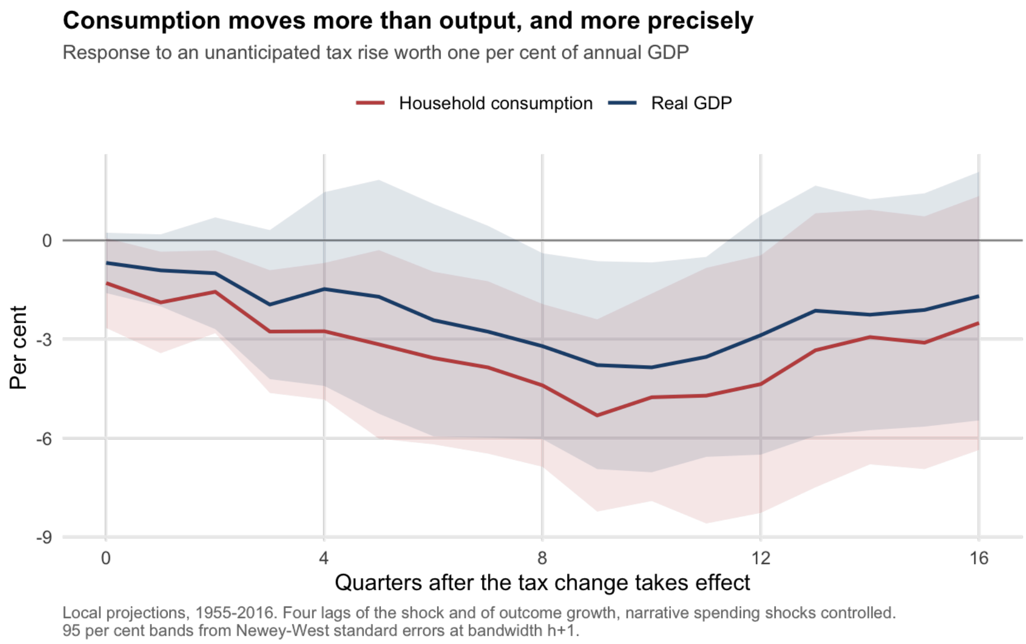

Figure 1: Consumption responds more than output, and on more horizons.

The stronger reason to prefer consumption is robustness. The published UK tax multiplier rests heavily on one Budget. In Cloyne's own 1955 to 2009 surprise series the June 1979 Budget, which raised VAT to 15 per cent, is the largest single observation at +1.596 per cent of GDP, and removing that quarter takes his aggregate output response from $t = -2.45$ with four significant horizons to $t = -1.46$ with none. The same is true of the series built here and of the 2025 revision. The output response does not survive it.

The consumption response does. Removing the same quarter leaves a trough of -3.78 with $t = -2.27$, so the sign, most of the magnitude and the significance all hold.

That single Budget is worth a further remark. It enters at +1.596 in the 2013 coding and at +0.377 in the 2025 revision, so the largest postwar observation in this literature moved by a factor of four between two vintages of the same authors' data. Any estimate that leans on it is leaning on a coding judgement, which is a reason to prefer estimates that do not.

4 The same tax rise costs twice as much in a slack economy

The state variable is an indicator for unemployment above its own seven-year backward moving average. It is backward-looking by construction, so it uses only information a policymaker had at the time, and it requires no output-gap estimate or filter. It puts 149 of the 248 usable quarters in slack.

Two separate estimates that differ are not evidence that the difference is real. Table 6

Table 5: Consumption response by the state of the economy

State	Trough	Quarter	95 per cent interval	t	Significant horizons
Economy has slack	**−7.31**	11	$[-9.55, -5.06]$	−6.38	**14 of 17**
Economy does not	−3.16	8	$[-6.53, +0.20]$	−1.84	4 of 17

The narrative spending control is identically zero within these subsamples and is dropped from them. Removing the largest single quarter in each state leaves the slack trough at −7.37 with $t = -4.32$.

therefore estimates the difference itself, by interacting the shock with the state indicator, so that the gap carries its own standard error.

Table 6: The difference between the two states, estimated directly

Quarters after the change	Slack minus tight	Standard error	t
4	−0.41	1.78	−0.23
8	−3.50	1.95	−1.80
9	−5.44	1.69	−3.22
11	**−8.25**	2.14	**−3.86**
12	−8.16	2.30	−3.54
16	−6.47	2.46	−2.63

Significant at eight of seventeen horizons. The gap opens from about the fifth quarter and is widest at eleven, which is also where the slack response troughs.

Figure 2 shows what the two states look like. The paths are indistinguishable for the first four quarters and separate after that: the slack response keeps falling and stays down, while the tight response troughs shallowly at two years and returns to zero.

Expressed the way a Chancellor would want it, and taking consumption at 61 per cent of GDP, a tax rise of one per cent of GDP is followed at its worst by a fall in household spending of about 4.5 per cent of GDP when there is slack, against about 2 per cent when there is not. The trough arrives at two to three years in both cases, so this is a claim about the medium run and not about the quarter of impact.

4.1 What is not shown

The multiplier does appear to fall as the pre-announcement regime takes hold, from a trough of −4.52 over 1955 to 1985 to −2.97 over 1986 to 2016. The late interval spans zero and the two overlap heavily, so this is a direction and not a finding, and it is reported here rather than in a table for that reason.

A tempting bridge to the companion paper also fails to carry weight. If tax changes are now scheduled years ahead, a Chancellor might be unable to see the state a measure will land in, which combined with the result above would be a first-order cost of pre-announcement. The correlation between slack at announcement and slack at implementation does fall with notice, from 0.995 for measures taking effect within 90 days to 0.806 for those taking more than a year, and the typical state moves by 0.74 points rather than 0.03. That is real and it is modest. Unemployment is persistent enough that even two years out a Chancellor mostly knows the state, and the claim should be made at that size.

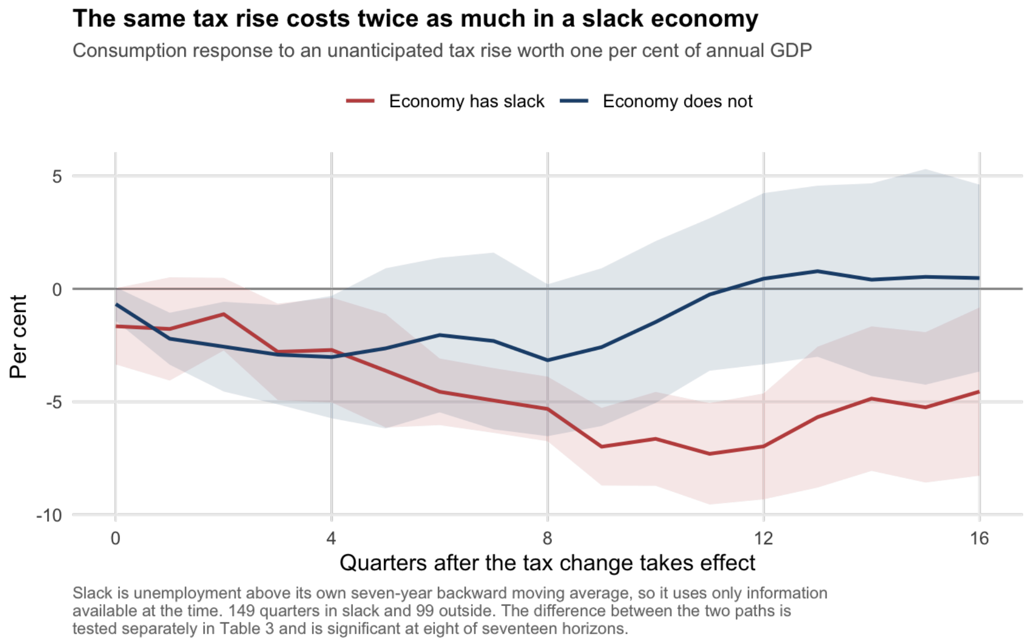

Figure 2: Consumption response by state, with 95 per cent bands.

5 What the record cannot support

Three questions were posed and abandoned. Each is reported here with the number that killed it, because a measured limit is more use to the next researcher than a caveat.

Which tax. Instrument-level multipliers look estimable on measure counts, with 449 income tax measures, 350 excise and 175 corporate. They are not estimable on variation. Eighty-four per cent of the absolute variation in the VAT surprise series sits in three quarters and the largest is June 1979; capital gains is 64 per cent concentrated and corporation tax 47 per cent. The resulting estimates are single episodes extrapolated to a shock worth one per cent of GDP, which for capital gains is a fifty standard deviation extrapolation, and they duly return implausible magnitudes. The UK narrative record cannot support a tax-by-tax decomposition.

Whether notice blunts the effect. The natural sequel to the companion paper is that pre-announced tax changes move consumption less, because households have already adjusted. The announcement-dated series returns no detectable response at any horizon. That null is uninformative: the minimum detectable effect of the design, the smallest true response it would find four times in five, is 10.5 per cent against a surprise response of 5.3. The test cannot distinguish no announcement effect from an announcement effect exactly as large as the surprise. Nothing should be concluded from it in either direction.

Who bears the cost. This is the subject of the next section.

6 Work in progress: the distributional half

This section describes work that is not finished. It is included because the question it asks is the one the paper most wants to answer, and because the route to answering it is now clear.

If a pound taken from a household with no buffer cuts spending by more than a pound taken from a household with savings, then the aggregate response above conceals a distribution, and two tax packages of the same size can have different costs depending on where they land. That is the mechanism Cloyne et al. (2025) and the heterogeneous-agent literature would predict, and it is the policy question behind the whole exercise.

6.1 What is already possible, and what it shows

The ONS Effects of Taxes and Benefits tables report, for every year from 1977, what each income decile pays in income tax, National Insurance, council tax, VAT and each duty. That gives the cash incidence of every instrument, and 1,036 measures carrying 83 per cent of gross value can be placed on it. Corporation tax, capital gains, inheritance and North Sea measures cannot be placed, because the ONS tables never allocate those to households.

The gradient is real and correctly ordered. The bottom five deciles bear 0.36 of duties, 0.31 of VAT, 0.19 of stamp duty, 0.16 of National Insurance and 0.15 of income tax. These are cash shares, so none reaches a half: richer households pay more of every tax in absolute terms. What the ordering measures is relative concentration, and duties lean more than twice as far down the distribution as income tax does. Figure 3 shows the series.

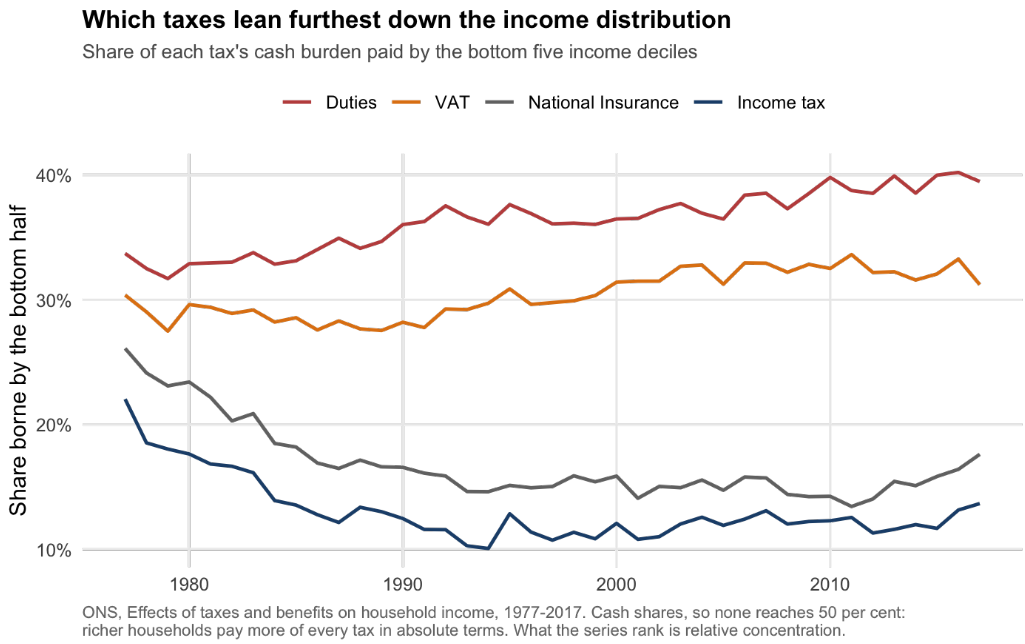

Figure 3: Share of each tax's cash burden borne by the bottom five income deciles.

Splitting each measure into a bottom-borne and a top-borne part and projecting consumption on each gives troughs of −14.3 and −5.1: a pound taken from the bottom half looks like it costs roughly three times the consumption of a pound taken from the top. That is the right sign and the expected ordering, and it is not a result. The bottom-borne interval runs from −31.3

to +2.7 and includes zero, the two series correlate at 0.93 because they are slices of the same measures rather than different ones, and only 15 to 36 per cent of any measure lands on the bottom half, so the per-pound coefficient extrapolates 24 standard deviations. Standardised to a one standard deviation shock the two responses are 0.60 and 0.65 per cent of consumption, which is to say indistinguishable.

The obstacle is structural rather than statistical. Slicing one shock cannot identify heterogeneity, because nothing in the data separates the slices. Identification requires either measures that differ in incidence, which is the instrument decomposition ruled out above, or households observed directly.

6.2 What the finished version requires

Household micro data. The Living Costs and Food Survey and its predecessor the Family Expenditure Survey record household expenditure from 1961 and would allow the consumption response to be estimated separately by position in the income or liquidity distribution, in the manner of the household-level fiscal transmission literature. The Wealth and Assets Survey, from 2006, is the only UK source carrying wealth and liquidity directly, but it begins exactly where the surprise series has lost four-fifths of its amplitude, so the long sample and the wealth measure cannot currently be had at once. All three are held by the UK Data Service under an End User Licence.

The specification is already written and the aggregate it decomposes is already estimated. What remains is data access, and until that is resolved this paper reports the aggregate consumption response and the incidence gradient separately, without joining them.

7 Conclusion

The timing of a tax rise is an economic decision and not only a political one. On UK data from 1955 to 2016, the same unanticipated tax rise is followed by roughly twice the fall in household consumption when unemployment is above its recent average as when it is below, and the difference is large enough, and estimated precisely enough, to matter for the choice of when to consolidate.

Two cautions belong with that. The result is about the medium run, with the gap opening after a year and widest at eleven quarters, so it says nothing about the quarter of impact. And it is a statement about the average tax rise, because the record cannot separate one instrument from another and cannot yet say whose spending falls.

The narrower methodological point may outlast the estimate. The published UK tax multiplier depends on a single Budget in June 1979 whose coded value moved by a factor of four between two vintages of the same dataset, and the output response does not survive its removal while the consumption response does. Anyone estimating fiscal multipliers on this record should check which of their results are statements about the British economy and which are statements about that one Budget.

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